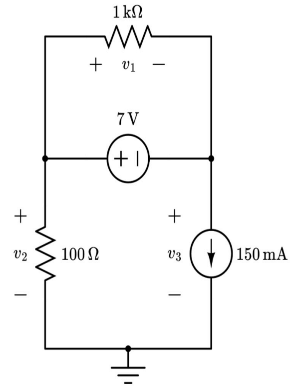
Practice Quiz Four: Node Voltage Analysis (Easy)

For the circuit shown below, use node voltage analysis to find the value of v_1, v_2, and v_3.

Suggested time limit: 10 minutes



Step #1 Constraint

$V_2 - V_3 = 7$

$V_1 = V_2 - V_3$

Step # 2 Supernode

$$\left(\frac{V_2 - 0}{100} + \frac{V_2 - V_3}{1000} + \frac{V_3 - V_2}{1000} + (150 \cdot 10^{-3}) = 0 \right) 1000$$

$10V_2 + V_2 - V_3 + V_3 - V_2 + 150 = 0$

$10V_2 = -150$

$V_2 = -15V$

Step # 3 Solve

$-15 - V_3 = 7$

$V_3 = -22V$

$V_1 = -15 - (-22)$

$V_1 = 7V$